

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 2

Year: 2024

Subject: 031021122 Electric Circuits II

Section: 5-6

Date: 10 January 2025

Time: 15.00-17.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 3 sections. There will be 11 pages (including this page) and they are worth 60 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

Now begin the test.

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**Cheating in the exam is considered an extremely serious offence
which will result in expulsion from the University.**

This exam paper is part of a set used to assess student abilities as follows:

		Part 1	Part 2	Part 3
CLO 1	Explain Capacitor and inductor in DC circuit, alternating current (AC) circuits, Sinusoidal alternating waveforms, sinusoidal voltage and current, phase relations, average value, effective (rms) value, basic R, L, and C elements in AC circuits, average power. The power triangle, power-factor correction, three phase generator, Y-connected generator, phase sequence, and star-delta system. Rectangular form, polar form, series and parallel AC circuits, series-parallel AC networks	√		
CLO 2	Identify the types of capacitor, inductor and resistor in AC networks.	√	√	
CLO 3	Analyze the series capacitor, inductor and resistor in AC networks, parallel capacitor, inductor and resistor in AC networks and series-parallel capacitor, inductor and resistor in AC networks using methods of analysis for AC networks		√	√
CLO 4	Design the series capacitor, inductor and resistor in AC networks, parallel capacitor, inductor and resistor in AC networks and series-parallel capacitor, inductor and resistor in AC networks using methods of analysis for AC networks		√	√

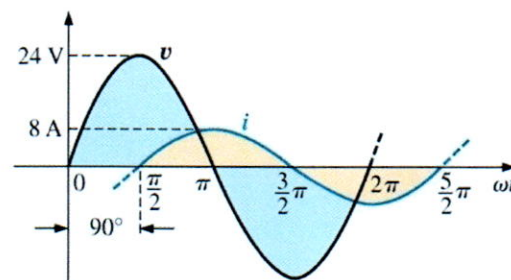
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Section 1 (20 points) True/False

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. The insulating material in capacitor can be made from air, glass, rubber, copper, plastic, etc.
2. In the discharging phase of a capacitor, the voltage of the capacitor decreases exponentially.
3. At transient state, the equivalent open circuit represents the behaviors of a capacitor.
4. In RC network, a steady state occurs when the switch is turned off for 5 time constants.
5. When discharging a capacitor, the current of the capacitor is approaching to zero.
6. The time constant for a capacitor circuit equals its capacitance multiplied by resistance.
7. In RL network, the energy will be fully stored before 5τ .
8. In real life, the inductor is usually made from copper. The copper coils therefore have no internal resistance in Ohm.
9. The energy storage of an inductor is direct proportional to the inductance value of the inductor.
10. An ideal inductor is a short circuit in DC circuit.
11. The current across an inductor lags the voltage through it by 90° .
12. The power factor of a resistive circuit is equal to zero.
13. Consider the total power equation $P_T = EI \cos \theta_T$, which is same to the power equation $P_T = I^2 R$.

14. The magnitude of polar form is in effective value.
15. The impedance of an inductor is the phasor.
16. A phasor diagram represents sinusoidal waveforms as rotating vectors in a plane, where the length of the phasor indicates the amplitude, and the angle represents the phase.
17. Two sinusoidal voltages $V_1 = 10 \angle 30^\circ$ V and $V_2 = 5 \angle -60^\circ$ are represented on a phasor diagram. The resultant voltage V_R is calculated as $V_R = V_1 + V_2 = 15$ V.
18. Given $R = 4 \, \Omega$, $X_L = 8 \, \Omega$, the total impedance is $8.94 \angle 63.43^\circ \, \Omega$.
19. Given a sinusoidal voltage source $V = 120$ V is applied across the circuit with $R = 4 \, \Omega$, and $X_L = 8 \, \Omega$, the current I and its phase angle with respect to the voltage source is $13.42 \angle -63.43^\circ$.
20. The voltage $V = 16.97 \angle 0^\circ$ Vrms and the current $I_{rms} = 5.66 \angle -90^\circ$ A and the current are shown in figure below.



Section 2 (20 points) Multiple choices

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. A sinusoidal voltage source $v(t)=50\sin(1000t)$ V is applied across a capacitor with a capacitance of $C=10\ \mu\text{F}$. Determine the capacitive reactance X_C and the peak current I_m through the capacitor.
 - (a) $X_C = 10\ \Omega$, $I_m = 5\text{A}$
 - (b) $X_C = 100\ \Omega$, $I_m = 0.5\text{A}$
 - (c) $X_C = 100\ \Omega$, $I_m = 5\text{A}$
 - (d) $X_C = 10\ \Omega$, $I_m = 0.5\text{A}$
2. A sinusoidal current $i(t)=5\sin(200t)$ A flows through an inductor with an inductance of $L=0.2\text{ H}$. Determine the inductive reactance X_L and the peak voltage V_m across the inductor.
 - (a) $X_L = 40\ \Omega$, $V_m = 200\text{V}$
 - (b) $X_L = 20\ \Omega$, $V_m = 100\text{V}$
 - (c) $X_L = 40\ \Omega$, $V_m = 100\text{V}$
 - (d) $X_L = 20\ \Omega$, $V_m = 200\text{V}$
3. Given $R=3.4\text{ k}\Omega$ and $C=3.3\ \mu\text{F}$, What is the time constant?
 - (a) 11.22 ms
 - (b) 6.6 ms
 - (c) 3.96 ms
 - (d) 3.4 s
4. What is the capacitor voltage equation at charging state?
 - (a) $(E/R)*e^{t/\tau}$ [V]
 - (b) $E*e^{-t/\tau}$ [V]
 - (c) $(E/R)*(1-e^{t/\tau})$ [V]
 - (d) $E*(1-e^{-t/\tau})$ [V]
5. Given $R=3.4\text{ k}\Omega$, $E=24\text{V}$ and $C=3.3\ \mu\text{F}$, What is the capacitor current equation at charging state?
 - (a) $7.06e^{t/\tau}$ [mA]
 - (b) $7.06e^{-t/\tau}$ [mA]
 - (c) $70.6e^{t/\tau}$ [mA]
 - (d) $70.6e^{-t/\tau}$ [mA]

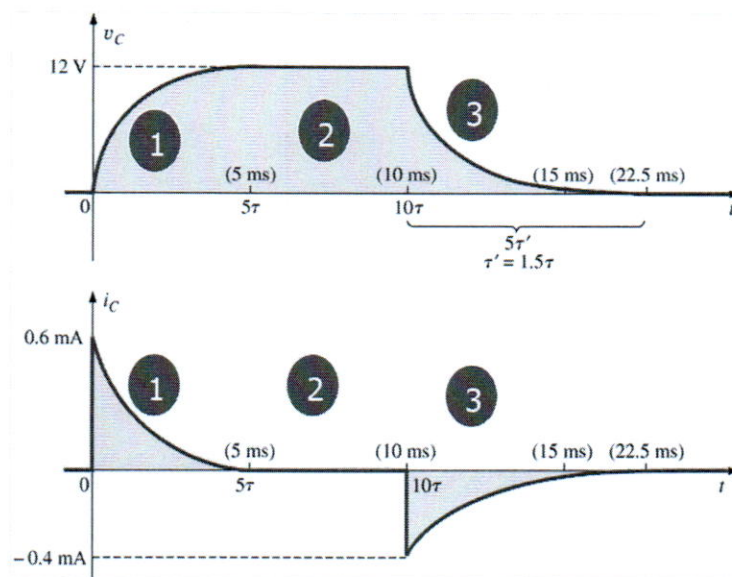


FIGURE 1

6. What is the voltage across the capacitor after 5τ during the charging phase from figure 1.
 (a) 6 V (b) 9 V (c) 11.3 V (d) 12 V
7. How much time is required for the capacitor to discharge from $V_C=12$ V to approximately 1.5 V during the discharging phase? from figure 1.
 (a) 5 ms (b) 10 ms (c) 15 ms (d) 22.5 ms
8. The voltage across a coil is $v(t)=100\sin(50t)$ V and the current is $i(t)=20\sin(50t-90^\circ)$ V. Which of these is the inductance of the coil?
 (a) 1 H (b) 100 mH
 (c) 100 H (d) 10 H
9. Find the capacitance of a parallel plate capacitor if the area of each plate is 0.075 m^2 and the distance between the plates is 1.77 mm. The dielectric is mica ($\epsilon_r = 5$) where $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$.
 (a) 1.04 pF (b) 1.04 nF (c) 1.874 pF (d) 1.88 nF
10. At what frequency does a $1 \mu\text{F}$ capacitor have a reactance of 100Ω ?
 (a) 15.9 kHz (b) 1.59 kHz (c) 159 Hz (d) 1.59 MHz
11. What is the inductive reactance at 80 Hz of a 1 mH inductor?

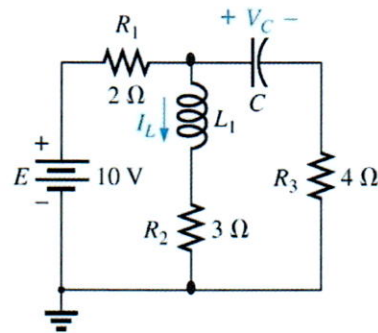
(a) $50 \, \Omega$ (b) $0.5 \, \Omega$ (c) $20 \, \Omega$ (d) $0.2 \, \Omega$ 

FIGURE 2

12. In steady state, what is the current through the resistor R_1 from figure 2?

(a) 1 A

(b) 2 A

(c) 4 A

(d) 5 A

13. What is the phasor form of voltage (v) in figure 3?

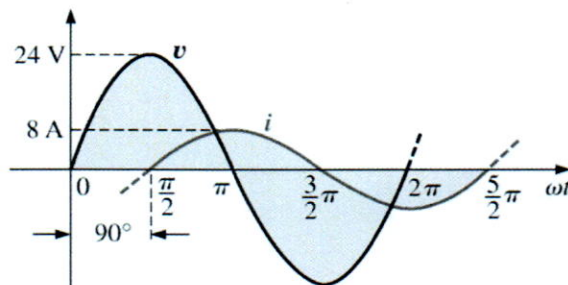


FIGURE 3

(a) $V = \frac{24}{\sqrt{2}} \angle 0^\circ$

(b) $V = \sqrt{2}(24) \angle 0^\circ$

(c) $V = \frac{24}{\sqrt{2}} \angle 90^\circ$

(d) $V = \sqrt{2}(24) \angle 90^\circ$

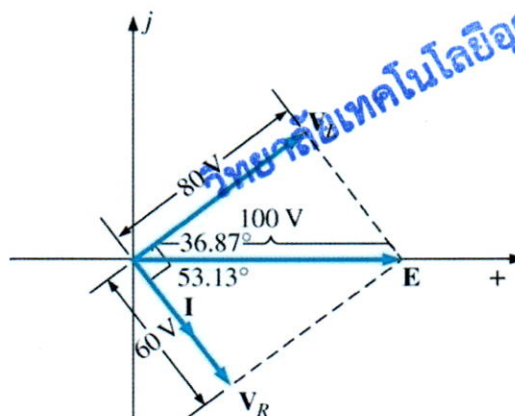


FIGURE 4

14. What is the time domain equation of voltage (v_L) from figure 4?

- (a) $v_L = \sqrt{2} \times 80 \sin(\omega t + 36.87^\circ)$ (b) $v_L = 80 \sin(\omega t + 36.87^\circ)$
 (c) $v_L = \sqrt{2} \times 80 \sin(\omega t - 36.87^\circ)$ (d) $v_L = 80 \sin(\omega t - 36.87^\circ)$

15. Find the peak to peak value of the waveform in Figure 5.

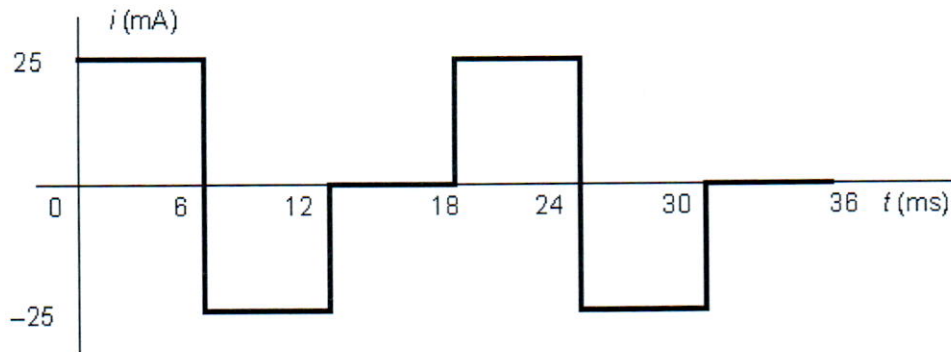


FIGURE 5

- (a) 50 mA (b) 12.5 mA (c) 25 mA (d) 20 mA
16. Using the waveform in Figure 5, find the period of the current.
 (a) 36 ms (b) 18 ms (c) 24 ms (d) 12 ms
17. Using the waveform in Figure 5, find the average value of the current.
 (a) 0 mA (b) 25 mA (c) 17.68 mA (d) 12 mA
18. What is the effective (rms) value of $v = 23 \sin 400t$?
 (a) 400 V (b) 23 V (c) 16.26 V (d) 11.5 V
19. Express the phasor $100 \angle 60^\circ$ as a sine wave with an angular velocity of 60 rad/s.
 (a) $141.4 \sin(60t + 60^\circ)$ (b) $100 \sin(377t + 60^\circ)$
 (c) $141.4 \sin(377t + 60^\circ)$ (d) $100 \sin(60t + 60^\circ)$
20. Which one of the following phasor domain expressions is equivalent to the time domain expression $50 \sin(100t + 15^\circ)$?
 (a) $70.7 \angle -15^\circ$ (b) $50 \angle 15^\circ$
 (c) $35.35 \angle 15^\circ$ (d) $70.7 \angle 15^\circ$

Section 3 (20 points) Calculation

Instruction: Derive the mathematical expression and answer of following problems.

- Find the total admittance and impedance of the circuit of Figure 6.

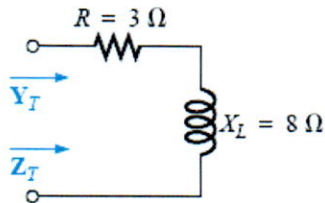
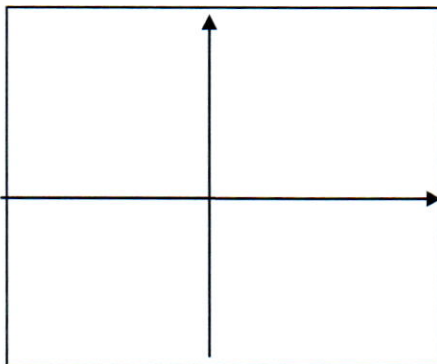


FIGURE 6

- Given a parallel RLC circuit has the following components: $R=10 \Omega$, $L=0.1 \text{ H}$ and $C=100 \mu\text{F}$, the applied voltage $V=100 \angle 0^\circ \text{ V (RMS)}$ and the frequency $f=50 \text{ Hz}$. Calculate the current I_R , I_L and I_C and draw the phasor diagram.

Sketch the phasor diagram.



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3. Calculate the currents I_1 and I_2 of Figure 7 in phasor form using the current divider rule.

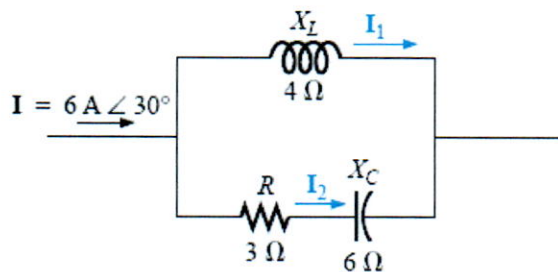


FIGURE 7

4. For the network of Figure 8.
- Find the total impedance of the circuit
 - Compute I_s , I_2 and V_c
 - Sketch the phasor diagram of I_s , I_2 and V_c

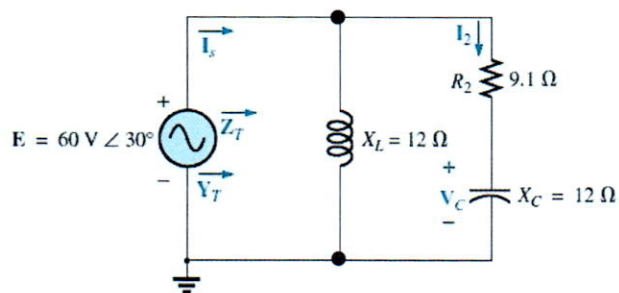
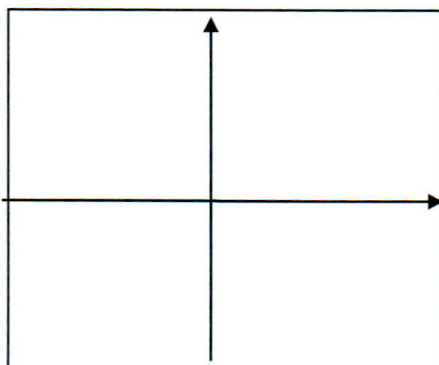


Figure 8

Sketch the phasor diagram.



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and Asst. Prof. Dr. Kittisak Phaebua

Answer Sheet

Name: _____ ID: _____

Subject: 031021122 Electric Circuits II

Section 1

	True	False
1.		
2.		
3.		
4.		
5.		
6.		
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Section 2

	a	b	c	d
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